Hybrid Retrieval for Retrieval-Augmented Generation Using Reciprocal Rank Fusion and Cross-Encoder Re-ranking

Kapil Kumar Gupta
Computer Science & Engineering Department
Amity University, Lucknow, Uttar Pradesh, India
kkgupta@lko.amity.edu

Arpita
Computer Science & Engineering Department
Amity University, Lucknow, Uttar Pradesh, India
arpita9601@gmail.com

Namrata Dhanda
Computer Science & Engineering Department
REC, Mirzapur, India
ndhanda510@gmail.com

***Abstract*— Not useful information or bad lists can stop the trust of systems that give answers. Therefore, excellent retrieval quality is a strong reason for success in retrieval-augmented generation (RAG). Many think that the quality of retrieval determines the correctness of the final answer. This document presents and evaluates a hybrid method of working for gathering supporting facts for the process of giving answers that need many steps. This work method combines the BM25 word-matching system with the deep meaning search using Sentence-BERT MiniLM embeddings and FAISS. Two different lists of information are merged by using Reciprocal Rank Fusion (RRF) and a Cross-Encoder is used to fix the order of the pieces of text. We conduct investigative tests on the full 7,405-question HotpotQA distractor validation split, providing each hard question requiring multiple steps with ~10 pieces of text and 2 supporting documents. We report Recall@1, Recall@3, Recall@5 and Mean Reciprocal Rank (MRR) to measure success of information gathering. The BM25 tool achieves 80.02% Recall@5 and 87.68% MRR, while MiniLM with FAISS reaches 82.88% and 89.24%. When the hybrid RRF system is used, these specific numbers are increased to 85.35% and 90.91%. The All-candidate Cross-Encoder re-ranking yields the best results for the lists, with Recall@1 45.89%, Recall@3 79.35%, Recall@5 86.73% and MRR 95.19%. The researchers find that word signals and meaning signals work well together and the application of computer brain re-ranking can greatly improve the rank of important facts. Our academic writing is now on the stage of getting information instead of making the last answer. The quality of retrieval provides a study for picking supporting facts that others can repeat for the process of giving answers that need many steps.**

Keywords—Retrieval-Augmented Generation, Hybrid Retrieval, BM25, Sentence-BERT, FAISS, Reciprocal Rank Fusion, Cross-Encoder, HotpotQA, Multi-Hop Question Answering.

Introduction

Large language models are successful in tasks requiring a lot of knowledge even if they have limited and old internal knowledge. Retrieval-Augmented Generation (RAG) solves the limitation because Retrieval-Augmented Generation (RAG) finds outside proof and provides that proof to a following system [1]. Many assume that the text generation stage of a RAG pipeline only works on information that the finding stage finds first. Therefore, the quality of searching is viewed as one of the key factors for the truth and the trust of answers [8], [11].

Older word-matching approaches such as BM25 are still useful because BM25 uses exact words and the frequency of words in papers. Instead of working in a word space, dense retrieval works in a space for meaning, so related texts can be found even if the words are not. Trained vector forms are useful for question answering, as demonstrated by Dense Passage Retrieval [2]. Sentence-BERT made the math for meaningful embeddings easy to use [3]. There is no one system that is better than the other because word systems work well for specific names, while dense systems are good at capturing meanings and similar ideas.

Mixed searching combines different kinds of signals. Reciprocal Rank Fusion (RRF) is selected because Reciprocal Rank Fusion (RRF) fuses lists without having to modify the scores first. Recent research on RAG mix investigates word searching, dense embeddings, and neural sorting [9], [14], [15] according to experts. Cross-Encoders, which look at a question and a text together to find the match, achieve a very high accuracy [4]. Computing cost is high, so the first search is performed with Cross-Encoders to re-sort lists.

We evaluate the mixed system on HotpotQA, a multi-hop question answering [5]. The setting with extra papers is good for analysis because there is a question shown with right and wrong papers. This complete group of data contains 7405 difficult questions . Each question has about 9.95 possible texts and two right papers . The proposed pipeline compares four setups: BM25, dense search with MiniLM and FAISS, mixed RRF and RRF with sorting at the end with Cross-Encoder.

The idea here is not to teach a new computer model or to show how the whole system answers. This paper separates the finding part because the paper wants to see if combining word matching, meaning search and neural scores helps the order of proof. The main contribution of this work is the comparison of the finding stage on the HotpotQA data according to experts. This work is supported in Recall@k and MRR scores and an analysis of the problems of the test setting .

I. Literature Review

Lewis et al. [1] introduced Retrieval-Augmented Generation for assignments requiring deep knowledge by connecting a parametric model with retrieved memory. Many scholars have said that external retrieval helps models to gain better knowledge during these knowledge intensive tasks . Dense Passage Retrieval (DPR) was proposed by Karpukhin et al. [2] as dense representations outperform sparse methods for question answering. Reimers and Gurevych [3] proposed efficient embeddings for cosine-similarity retrieval using Sentence-BERT. Nogueira and Cho [4] showed that BERT-based Cross-Encoders can improve the ranking of passages.

The evaluation is performed on HotpotQA [5] to measure the multi-hop reasoning with evidence of systems. Gold evidence is interspersed with unrelated passages here, because the distractor setting permits controlled experiments. In their work on BEIR, Thakur et al. [6] noted that the retrieval performance varies across the different datasets considered. Gao et al. [8] explored the transition of the system from simple forms to modular Retrieval-Augmented Generation architectures. Optimization was stressed by Gao

et al. [8] so the separate parts of the system, retrieval and generation, need to be verified separately.

Today, scholars try to improve the retrieval control and the quality of evidence. RAG-Fusion, a combination of rankings with RRF, was studied by Rackauckas [9]. Asai et al. [10] argue that Self-RAG allows models to flexibly inspect and assess evidence as claimed by experts. Yan et al. [11] developed Corrective RAG to make sure that the system checks the quality of the retrieval before starting the generation step. Retrieval errors often damage the final output, so corrective actions are taken when the evidence is not strong.

The methods of testing have changed to explore specific parts of the system. RAGAS [7] proposed metrics to check retrieval and generation without the need for humans to annotate everything. Many believe that diagnostic measures are essential for RAG components and this is why RAGChecker [12] was introduced. FlashRAG [13] provided a toolkit and resources for researchers to compare different methods. RankRAG [14] integrated the ranking of context and generation because the quality of ranking affects the performance of Retrieval-Augmented Generation. Caraman and Silaghi [15] used a three-stage system for SemEval-2026 based on query rewriting and hybrid methods.

Existing studies have a gap in that retrieval systems require many different signals. Before attributing the improvements to the entire pipeline, researchers should assess the contribution of each stage separately. It has been observed that improvements are applied to the entire pipeline before the parts are tested. The present work explores this gap by considering a controlled HotpotQA distractor setting. We evaluate sparse retrieval and dense retrieval together with rank fusion and Cross-Encoder ranking on 7,405 questions since this method gives a clear measurement.

II. METHODOLOGY

1. Dataset and Experimental Setting

These experiments are conducted on the HotpotQA distractor validation set [5]. HotpotQA is a multi-step query-response evaluation dataset built on Wikipedia, which contains questions, answers, supporting context information, and factuality labels. Experts say it has the required evidence and misleading writing sections of the distractor arrangement. The validation file contains 7,405 instances, 5,918 bridge questions and 1,487 comparison questions. Although these experiments use the validation examples, each validation example has a hard label. After the change, each question has a mean of 9.95 possible text sections and 2.00 useful evidence sections. We perform regulated ranking tests for each question as these experiments do not search the entire Wikipedia.

Dataset Characteristic	Value
Dataset	HotpotQA distractor validation
Questions	7,405
Bridge questions	5,918
Comparison questions	1,487
Difficulty	Hard
Avg. candidate passages/question	9.95
Avg. relevant passages/question	2.00

Table I. Characteristics of the experimental dataset.

2. Text Preparation

Each inquiry is grouped together with the heading and phrases that belong to each contextual section into a single potential record. We have seen that BM25 uses a basic small-letter linguistic segmenter because the system requires simple patterns. The authors of the paper note that the pretrained sentence-transformers/all-MiniLM-L6-v2 model for dense retrieval directly converts the same heading-and-text representation. The names of assisting records supplied by HotpotQA are retained as markers of importance for the evaluation process, while evaluating the system logic. The names of assisting records remain unchanged, because during the assessment process we need accuracy.

3. Sparse Retrieval with BM25

Researchers state that BM25 evaluates potential segments through the saturation of term occurrence, the frequency of inverse documents, and the normalization of text length. This lexical foundation is utilized when the ranking of potential segments is performed by BM25. Because BM25 gives a score to every potential segment for each inquiry, a list that descends in order is created. BM25 identifies the exact word matches between the inquiry terms and the written segments.

4. Dense Retrieval with MiniLM and FAISS

The all-MiniLM-L6-v2 Sentence-Transformer is utilized by Dense Retrieval to transform the inquiry and the potential segments into code. Even though L2-normalization is applied to the embeddings, the FAISS IndexFlatIP is employed so that the search for inner products remains equal to the ranking of cosine-similarity. Experts claim that Dense Retrieval finds evidence with a related meaning even when the inquiry and the supporting segment do not have matching surface words. Dense Retrieval performs this action because semantic connections are prioritized over simple vocabulary matches.

5. Reciprocal Rank Fusion

Individuals who study retrieval combine the BM25 and dense rankings with the method of Reciprocal Rank Fusion. Regarding a candidate document d, the fusion score is established by the equation:

$$RRF(d) = 1/(k + r_BM25(d)) + 1/(k + r_Dense(d)), \quad k = 60$$

In this formula, r_BM25(d) and r_Dense(d) represent the rank positions that the two searching systems produce. Because Reciprocal Rank Fusion uses ranks instead of raw BM25 and cosine scores that cannot be compared, no extra score-normalization step is needed. People observe that Reciprocal Rank Fusion functions well when rankings are joined this way.

6. Cross-Encoder Re-ranking

A pretrained cross-encoder/ms-marco-MiniLM-L-6-v2 model is utilized for the final ranking scientific test. Since every inquiry-passage pair is jointly encoded, a relevance score is assigned to each pair before candidates are reordered. During the final scientific test that was documented, Cross-Encoder Re-ranking provides scores for all available distractor candidates which are approximately ten for every inquiry. Experts claim that the last row of data must be viewed as all-candidate neural re-ranking that follows the hybrid retrieval scientific test because this is not a large-corpus retrieval result. Such a distinction remains significant while Cross-Encoder Re-ranking prevents overclaiming the functioning purpose of RRF when the re-ranker sees the complete candidate set.

7. Evaluation Metrics

Scientific experts measure retrieval effectiveness by using Recall@1, Recall@3, Recall@5, and Mean Reciprocal Rank (MRR). Many people believe that Recall@k represents the fraction of relevant supporting passages recovered within the first k ranks when these are averaged across inquiries. The mean reciprocal rank of the first relevant passage is defined as MRR. Since every HotpotQA example contains two relevant supporting documents, the Recall@1 has a theoretical maximum of about 0.5 in this setting. Evaluation Metrics must not be viewed as classification accuracy even though the numerical data appears as fractions. It has been observed that Evaluation Metrics provide a clear view of how well the system finds information.

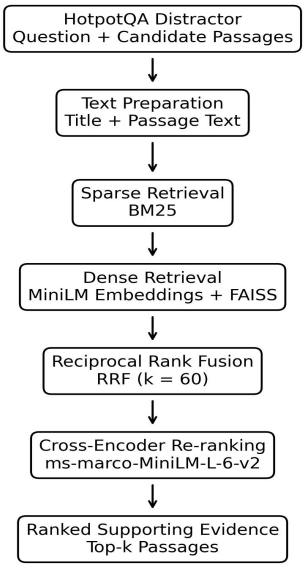


Fig. 1. Experimental retrieval pipeline.

8. Algorithmic Steps

The HotpotQA distractor validation instances are then given to researchers to obtain the specific query items, text segments, and supporting-document identifying markers. To clarify this, a complete named heading-plus-segment representation is constructed for each possible choice. The candidate candidates are tokenized and ranked according to BM25 technique. Many people believe this early organization helps the search process. The retrieval is more accurate as these potential selections are properly organized.

The system encodes the question item and candidate choices with MiniLM and ranks the dense vectors with cosine-equivalent FAISS inner-product search . The system fuses BM25 and dense rank positions using RRF with k=60. It has been found that combining these techniques is more effective. The Cross-Encoder is used to score pairs of query item-segments, enabling the system to reorder possible choices. These potential selections are carefully managed to identify the best matches.

The ranked named headings are compared with the HotpotQA supported named headings to check the accuracy of the model. Experts say that this comparison is the most important step in the work. Once the comparison is complete, the Recall@1, Recall@3, Recall@5 and MRR are summarized for all 7,405 inquiry items. The final statistics give a good overview of the system, because the performance is evaluated on 7405 inquiry items. This HotpotQA evaluation illustrates the true power of the retrival architecture.

III. Result and Discussion

1. Overall Retrieval Results

Measurement findings concerning the retrieval results are displayed in Table II. It has been observed that the ranking metrics grow better because every following retrieval stage happens in order. While MiniLM + FAISS makes all four metrics higher, BM25 offers a powerful word-based starting point. Efficiency is increased by RRF when RRF joins the different sparse and dense rankings that help each other. Numbers that were recorded reached the biggest amount when the all-candidate Cross-Encoder experiment was performed.

Method	Recall@1	Recall@3	Recall@5	MRR
BM25	0.399932	0.692302	0.800203	0.876774
MiniLM + FAISS	0.412694	0.722012	0.828832	0.892443
Hybrid RRF	0.424375	0.741053	0.853545	0.909089
RRF + Cross-Encoder*	0.458879	0.793518	0.867319	0.951911

Table II. Retrieval performance on 7,405 HotpotQA distractor validation questions.

When comparison is made with BM25, the final neural ranking experiment improves Recall@1 by 5.89 percentage points, Recall@3 by 10.12 points, Recall@5 by 6.71 points, and MRR by 7.51 points. The highest numeric growth is found within Recall@3. Experts claim that the combined retrieval and relevance-scoring stages function with very high success because the neural ranking experiment places helpful proof into the small context window that a downstream RAG system usually uses. Evidence is shifted into the limited space by the retrieval and relevance-scoring stages because this method makes the data more reachable for the machine. Because the neural ranking experiment is precise, the retrieval and relevance-scoring stages manage the information flow effectively.

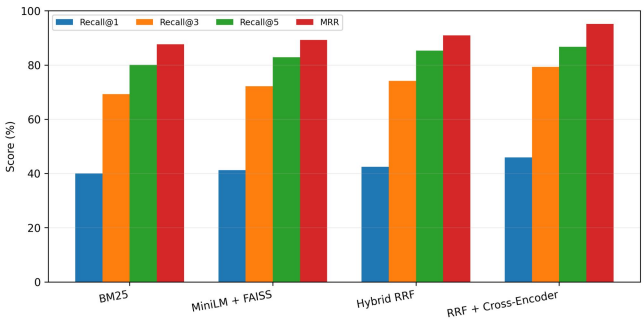


Fig. 2. Comparison of retrieval metrics across the four experimental configurations.

2. *Input and Output Example*

One HotpotQA comparison question is shown below so that people can see the question. Experts claim that the correct answer is kept only as data for the collection because this investigation measures evidence retrieval instead of checking if the created response is right. Since evidence retrieval is the priority, the accuracy of the produced outcome remains a secondary matter.

Input question	Were Scott Derrickson and Ed Wood of the same nationality?
Ground-truth answer	yes
Relevant supporting titles	Scott Derrickson; Ed Wood
Candidate context	Approximately ten passages, including Ed Wood, Scott Derrickson, Ed Wood (film), Adam Collis, Conrad Brooks, Sinister (film), and other distractors.

Fig. 3. Example retrieval input derived from HotpotQA.

In this specific situation, Ed Wood was placed at rank 1 by dense retrieval, while Scott Derrickson was positioned at rank 6. This distance demonstrates the difficulty which exists when a multi-hop comparison requires finding two different pieces of proof. It has been observed that gathering all necessary facts remains a difficult task for automated systems. When the Cross-Encoder scoring of the candidate group was completed, the two supporting documents were moved to the first two positions.

Rank	Passage title	Relevant?
1	Scott Derrickson	Yes
2	Ed Wood	Yes
3	Ed Wood (film)	No
4	Adam Collis	No
5	Conrad Brooks	No
6	Doctor Strange (2016 film)	No
7	Sinister (film)	No
8	Deliver Us from Evil (2014 film)	No
9	Tyler Bates	No
10	Woodson, Arkansas	No

Fig. 4. Example output ranking after Cross-Encoder re-ranking.

3. *Discussion*

The BM25 result shows that lexical matching is still capable in the HotpotQA distractor setting. An MRR of 0.8768 means that at least one supporting passage is often ranked near the top of the list. This shows the added value of semantic similarity to improve the dense MiniLM retrieval, raising MRR to 0.8924 and Recall@5 to 0.8288. This is consistent with previous dense-retrieval academic work [2], [3] that many people believe.

RRF improves all the metrics we measure without training a fusion model. The increase of Recall@5 from 0.8288 to 0.8535 suggests that sparse and dense systems have somewhat overlapping errors. RRF is purely based on rank positions according to the experts . Therefore, the different numeric scales of BM25 and cosine-similarity scores do not affect the method . This feature makes RRF a functional hybridization method and the logic agrees with recent hybrid RAG systems [9], [15]

The Cross-Encoder achieves the strongest ordering with MRR 0.9519. This is because the joint query-passage encoding allows a deeper interaction between tokens than the independent bi-encoder embeddings [4]. Fig. 4 shows the pattern of moving both required documents to ranks 1 and 2 by the system. The final experiment does re-rank about ten distractor candidates, so this test has to be considered as a controlled neural ranking result. This data does not constitute evidence of the same efficacy on a full Wikipedia corpus.

Another interpretation is in terms of Recall@1. Since there are two relevant documents for each question, retrieving one relevant document at rank 1 increases the recall by about 0.5. Hence, the Recall@1 of 0.4589 is near the maximum value achievable under this metric, but it does not mean that the answers are correct 45.89% of the time. As many observers have noted, MRR measures the position of the first relevant document. The metric does not show that both pieces of multi-hop evidence were retrieved. MRR also does not demonstrate that a downstream language model would answer correctly.

The results are consistent with the main hypothesis that retrieval excellence can be improved by combining lexical, semantic and neural relevance cues. At the same time the answer generation begins after the academic examination ends. This separation follows recent proposals that retrieval and generation components should be diagnosed separately by researchers [7], [12]. In the future, the researchers suggested an end-to-end experiment should use a fixed generator and measure answer Exact Match and token-level F1. An experiment on the full corpus should also verify if gains in retrieval are maintained when candidates are not pre-provided by the distractor benchmark.

IV. Conclusion And Future Works

This research paper examined a hybrid retrieval architecture which experts planned for the evidence-selection stage of Retrieval-Augmented Generation. Systematic assessments were performed on all 7,405 inquiries in the HotpotQA

distractor validation split. These assessments showed steady gains when the method moved from BM25 to dense MiniLM retrieval, and then progressed to RRF fusion and neural Cross-Encoder ranking. BM25 obtained Recall@5 of 0.8002 and MRR of 0.8768, while hybrid RRF increased these values to 0.8535 and 0.9091. It has been observed that all-candidate Cross-Encoder re-ranking obtained the most effective determined sequence with Recall@5 of 0.8673 and MRR of 0.9519. The gathered results show the combined strength of lexical and semantic retrieval because the significance of query-passage interaction is high during re-ranking.

The specific investigation is confined to the HotpotQA distractor candidate collection and the hybrid retrieval architecture does not assess answer generation. Experts claim that upcoming research activities should examine RRF-selected top-k Cross-Encoder re-ranking in a large shared corpus. A description regarding the joint recovery of both supporting written materials must be finished while researchers calculate retrieval waiting time. The hybrid retrieval architecture must be linked to a fixed open-source creation system. End-to-end assessment using Exact Match, token F1, faithfulness, and context relevance would provide a more complete assessment of RAG performance. Extra trials on heterogeneous benchmarks such as BEIR would also test cross-domain broad application. These extensions can make the current retrieval investigation a complete, reproducible RAG assessment structure.

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